

STA2020 Lab 1: R and RStudio

This lab will test your understanding of the basics of R and RStudio. Note that where words are in italics or included between quotation marks (or both) it requires input from you e.g. *"your name"* means that you would type your name. If you are working on your own laptop/PC, you need to have installed R and RStudio before attempting to do this lab.

Instructions:

- 1) Make sure that you have downloaded the dataset 'lab1.csv' and the pdf 'R and RStudio – lab 1'
- 2) Save it to your desktop or downloads folder.
- 3) Make sure you have opened RStudio.
- 4) Set your working directory to wherever you have saved the lab1.csv file by using: Session – Set working directory – Choose directory and navigate to wherever you have saved the dataset for this lab.
- 5) Insert a new R Script, using File – New File – R Script OR by clicking the white page with the green + button and selecting R Script
- 6) Type the following in the first few lines of your RScript:

```
# My name is "your name"  
# This is my first R and RStudio lab  
# Todays date is "todays date"
```

- 7) Import the lab1.csv dataset into RStudio by clicking on it and selecting 'Import dataset' and then selecting 'Import'
- 8) Type **attach(lab1)** into your script and then select 'Run' to attach the dataset to your workspace in RStudio
- 9) Type **mean(domestic)** into your script and then select 'Run' to compute the mean of domestic US profits
- 10) Type **median(nonfin)** into your script and then select 'Run' to compute the median of nonfinancial US profits
- 11) Type **var(corpprof)** into your script and then select 'Run' to compute the variance of adjusted corporate US profits
- 12) Type **hist(restworld)** into your script and then select 'Run' to plot a histogram of the rest of the world's profits
- 13) Type **summary(nonfin)** into your script and then select 'Run' to compute a 6-number summary of the nonfinancial US profits
- 14) Type **plot(year, corpprof)** into your script and then select 'Run' to plot the corporate US profits against year
- 15) Type **boxplot(domestic)** into your script and then select 'Run' to plot a boxplot of the domestic US profits